Validation criteria that exclude their own mechanism: six failure modes of voxel damage operators applied to Ni–YSZ tomograms

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Abstract

A simulation can satisfy every validity check its authors impose and still move the microstructure away from the physics it represents. We show this for voxel-scale damage operators on segmented tomograms, a standard way to simulate degradation in solid oxide cell electrodes, and say what to check instead.

A coarsening rule invites one obvious criterion: that it reduce specific surface area monotonically. We pre-registered exactly that. On a 6-connected lattice a single-voxel swap changes exposed area by exactly $2[\mathrm{nb}(a) - \mathrm{nb}(b)]$, so the criterion is a comparison of neighbour counts, and it leaves a large null space: four in five accepted moves change the area by exactly zero and are unpriced. On three FIB-SEM reconstructed Ni–YSZ anodes the operator satisfies the criterion in full, area falling monotonically at exact volume conservation, while triple-phase-boundary density rises three- to fivefold. The inflation is not diffuse: re-applying only the moves that touch the nickel–zirconia contact reproduces all of it, every other move giving $\times 1.04$. Strict inequality reduces the artifact without removing it. Surface area is the wrong invariant to validate a coarsening rule against.

The same criterion excludes Rayleigh-type neck break-up, which needs a transient area increase. No operator respecting it thinned a neck; the one that did violated it at its first step.

Four further modes are measured, each invisible on synthetic structures: largest-component pruning makes the spanning fraction identically unity, so the operator rewrites the denominator of the metric judging it; gates requiring perfect pristine connectivity misrepresent electrodes carrying 1.2 % to 11.2 % disconnected nickel; a jittered lattice cuts at one cross-section with zero variance where real networks fail at 0.5 % to 3 % of throats. One check costs a line: an operator whose TPB retention exceeds unity is roughening, not coarsening.

Keywords: solid oxide cell, Ni–YSZ anode, FIB-SEM tomography, percolation, triple-phase boundary, microstructural degradation

1. Introduction

Nickel–yttria-stabilised-zirconia cermet anodes lose performance in service through microstructural change. Nickel coarsens and can lose the percolating network that carries electronic current [1]; the ceramic scaffold fractures under redox cycling [2]; and the triple-phase boundary (TPB), the one-dimensional locus where nickel, zirconia and pore meet and where the electrochemical reaction proceeds, is lost as the nickel agglomerates [3]. All three are geometric processes acting on a microstructure that FIB-SEM and X-ray nano-tomography now resolve routinely, which makes a particular computational strategy very attractive: take a segmented tomogram, apply a damage operator to the label field, and track how connectivity and TPB evolve.

The strategy is attractive because it is cheap and because it acts on the same data structure the experimentalist already has. It is also, as far as we can tell, less examined than the continuum alternative. Phase-field treatments of Ni coarsening in this system have been developed and progressively validated over more than a decade, from the early three-phase simulations on experimentally acquired anode functional layers [4], through validation against time-dependent ex-situ tomography of one specimen imaged pristine and after annealing [5], to quantitative simulations initialised on ptychographic nano-tomograms with measured thermophysical parameters [6] and multiphase-field models coupled to transmission-line performance estimates on FIB-SEM reconstructions [7]. Discrete methods have been applied on the fabrication side, where a Potts kinetic Monte Carlo model of NiO–YSZ sintering was calibrated against FIB-SEM microstructure and validated on volume fraction, specific surface area and triple-phase boundary density [8]. Carrying that machinery over to degradation is less an open gap than an obvious next step, since it is the same free-energy formulation run under different driving conditions.

What is less examined is what happens when a discrete operator is instead assembled for the degradation problem directly, and how the conventions that surround such an

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operator interact with each other: the connectivity metric, the qualification gate on the test structure, and the validity criterion on the operator itself. We did not adopt the established formulation, and this paper is in part an account of what that cost.

This paper is that account, and it is negative in its first-order result. We set out to identify which damage mechanism reproduces a measured degradation signature in three Ni–YSZ anodes, and we did not find one. One operator reached a mechanism verdict on real tomograms, and reversed the measured ordering rather than reproducing it. Three more were run to a conclusion on a synthetic platform that we then had to disqualify. Four further routes closed before any mechanism test was reached, two because the operator failed its own validity gate and two because the data cannot pose the question. Along the way six specific ways in which this class of simulation silently misleads were identified and measured. We think the second finding is the more useful one, and the paper is organised around it, but the two are not separable: the failure modes were found by running the programme that failed, and they are reported in the order in which the work forced us to confront them.

The experimental target is unusually sharp and runs against intuition. Across the anode series of Pecho and co-workers [9, 10] the finest microstructure retains nickel percolation *worst* while retaining TPB *best* (Fig. 1, Table 1). A coarsening argument predicts the first half: by Herring's scaling laws the time to produce a geometrically similar change scales as a power of the particle size [11], so the finest structure should coarsen fastest and lose its network first. It does not obviously predict the second half, in which the same anode preserves four fifths of its reaction sites.

Two cautions travel with that target throughout this paper, and both follow from how the data were acquired rather than from how they were analysed. Medium versus coarse is not resolved on the nickel side: the ordering flips depending on whether percolation is defined by two-face spanning or by reachability from one face. And the published TPB series is not monotone, since coarse (0.608) exceeds medium (0.586). Only the two-level statements, *fine retains nickel percolation worst* and *fine retains TPB best*, are defensible, and those are the only claims we test against.

2. Materials and methods

2.1. Tomograms

Six segmented FIB-SEM stacks were used: three anodes, designated fine, medium and coarse by mean nickel particle diameter, in pristine and post-redox states. Stacks are 0.48 to 1.11 gigavoxels at pitches of 19.53 nm to 29.14 nm [9, 10]. Phase labelling was verified against the published volume fractions, with a worst deviation of 0.20 % across eighteen values.

Damage operators were applied to cubic regions of interest rather than to whole stacks, for tractability. Three non-nested regions were taken from each anode, at 8 µm for fine and medium. Coarse required 12 µm to reach the pre-registered floor of 150 graph nodes, and 12 µm is also the largest cube the coarse stack admits, since it is only 13.3 µm deep. Region size was frozen before any damage was applied and is not varied anywhere in this work; the resulting single-scale limitation is restated in Section 4.2.

The pristine and degraded stacks are different specimens (Rx36/Rx37/Rx38 against Rx41-1/Rx41-2/Rx41-3), not one volume imaged twice. Every retention magnitude is therefore confounded with specimen-to-specimen variation, and no paired before-and-after quantity is computed anywhere in this work. This is the principal reason all comparisons here are ordinal.

2.2. Percolation and connectivity

Phase connectivity uses 6-connectivity with free, non-periodic boundaries. A cluster percolates if a single connected component reaches both extreme slices of the transport axis; P_{span} is the fraction of phase voxels in such a cluster, and P_{reach} the fraction reachable from one face. The implementation recovers the simple-cubic site-percolation threshold as 0.31218 after finite-size extrapolation, against a reference value of 0.3116077(2) [12], a deviation of 6×10^{-4}.

2.3. Network extraction

Particle networks were extracted by watershed partitioning of the Euclidean distance transform of the solid phase [13]. Nodes are nickel chambers; edges are throats carrying a measured inscribed diameter and cross-sectional area. A defect in the library's default parallel dispatch was found and corrected during this work; its quantified effect on the earliest extractions is smaller than the between-region spread and changes no conclusion reported here.

2.4. TPB estimation

TPB density is the total length of edges shared by three differently labelled voxels, per unit volume. Gathering the four voxels around an edge with a circular shift wraps at the domain faces and manufactures triple lines; on an analytic single-triple-line test case the wrapping implementation over-counted by exactly a factor of four. The corrected estimator restricts the gather to interior edges. Estimates of TPB length from voxelised data carry a known systematic error whose size depends on the calculation rule and on the resolution [14, 15]; we therefore report TPB as a ratio to the pristine value of the same region wherever a conclusion depends on it, and quote absolute densities only for comparison against published values.

2.5. Damage operators

Two operator families were carried onto real tomograms. The first is stochastic surface erosion: at each round, every nickel voxel on the phase boundary is removed independently with probability $p_{\text{erode}} = 0.35$, uniformly over surface sites with no curvature weighting, after which the largest connected component is retained. A variant carrying an oxidative dilation step was tested and rejected on validity grounds (Section 3.3). The second is volume-conserving coarsening, in which a nickel voxel is removed from one site and added at another in the same round, so that the phase volume is exactly preserved. Three move-selection rules were implemented, all pre-registered. The first ranks sites by a local nickel-density proxy for curvature and was run on synthetic structures; the second ranks sites by a curvature proxy on an analytic test body; the third accepts a move only when it does not increase exposed area.

Three further operators were run on a synthetic platform whose nickel phase is a jittered regular lattice of spheres joined by explicit necks and whose YSZ phase is a particulate generator with an adjustable sintering yield. Initial yields were set from simple-cubic bond-percolation priors [16] and then recalibrated once, before any damage, against the measured connectivity of the generator's own face-centred-cubic contact graph. The three operators were: widening the lower tail of the neck-size distribution at conserved nickel loading, severing every throat in the lower quartile of the size distribution, and progressive fracture of sintered YSZ contacts. A cumulative-strain YSZ fracture operator was run on real tomograms. Definitions, frozen parameters and unit tests are in the repository.

The area-decreasing swap requires two specifications that turn out to change its behaviour, and both are recorded here because neither is a free choice after the fact. First, application is *sequential*: one move, neighbour field recomputed, repeat. Equation (2) is exact for a single swap against a current field, so a batch ranked once against a stale field has no guarantee, and in practice raises the area it is meant to lower. Sequential application on a 67-megavoxel region is only tractable with incremental updates, since a swap changes the neighbour count of just the twelve voxels adjacent to its two endpoints; candidates are held in seven buckets indexed by that count. Second, ties are broken *uniformly at random* among moves of identical ΔA, from a seeded generator. This is part of the rule, not an implementation detail: area-neutral moves dominate, so the achieved area reduction depends on which of many equal-cost moves is taken. At a matched budget of 1220 moves on the test body, a last-in-first-out order reaches $\Delta S_{\text{spec}} = -0.00024$ where the frozen randomised policy reaches -0.00955, a factor of 40 (`o7_tiebreak_sensitivity.py`).

Counting convention. The neighbour count nb in Eq. (2) excludes the voxel itself. This is worth stating because the standard 6-connectivity structuring element in common image-processing libraries includes the centre, and convolving with it returns $\mathrm{nb}+1$ on a nickel site against nb on a pore site. Any predicate comparing the two populations is then biased by one, which silently forbids every area-neutral move.

2.6. Outcome metric

Following the discovery described in Section 3.3 we adopted

$$R_{\text{Ni}} = \frac{\text{spanning-cluster nickel voxels}}{\text{pristine nickel voxels}}, \qquad (1)$$

which is invariant to the deletion of isolated material, monotone non-increasing under any removal operator, and directly comparable to measured retention. Its mandatory sanity check, that R_{Ni} evaluated on the undamaged structure equals the pristine spanning fraction, holds to 0.00×10^0 in all three anodes.

2.7. Statistics and pre-registration

There are three anodes. No p-values are reported anywhere in this work, and none should be computed from three points. Where correlations appear they are Spearman rank coefficients with leave-one-out sign checks against a threshold of $|\rho| \geq 0.6$ fixed in advance. Region-to-region and seed-to-seed spread is reported as a range rather than as an interval with a coverage claim.

The unit those correlations are computed over is the individual bisection, not the anode: three anodes × three regions × three damage seeds × two thresholds, i.e. 54 rows in `c1real_results.csv`, 27 per threshold. This is worth stating explicitly, because a rank coefficient over three points could only take the values $\{0, \pm 0.5, \pm 1\}$ and a threshold of 0.6 would then be equivalent to demanding $|\rho| = 1$. It is not that test. The corresponding caution is the opposite one: the 54 rows are nested within nine regions and three anodes and are therefore not independent, so the effective sample size is well below 54 and the coefficients are reported as descriptive rank checks with no inferential claim attached.

Each phase of the work was pre-registered in a version-controlled document committed before the run it governed, specifying the operator definition, frozen parameters, validity gates, decision rules and failure branches. Where a pre-registered rule later proved ill-chosen it was honoured and the defect reported, rather than amended after the fact. Two consequences of that policy appear below as reported nulls that a re-picked parameter would have avoided.

3. Results

3.1. The target

Table 1 and Fig. 1 give the outcome. Our absolute values track the published ones closely: P_{reach} matches the published P to within 0.1 % to 7.4 % on all six stacks. The one fact that survives every definition is that the fine anode retains nickel percolation worst, and it survives because the source papers state it independently. The fine anode is simultaneously the best TPB retainer, which is what makes the target interesting: whatever mechanism destroys the nickel network in the finest structure leaves most of its reaction sites intact.

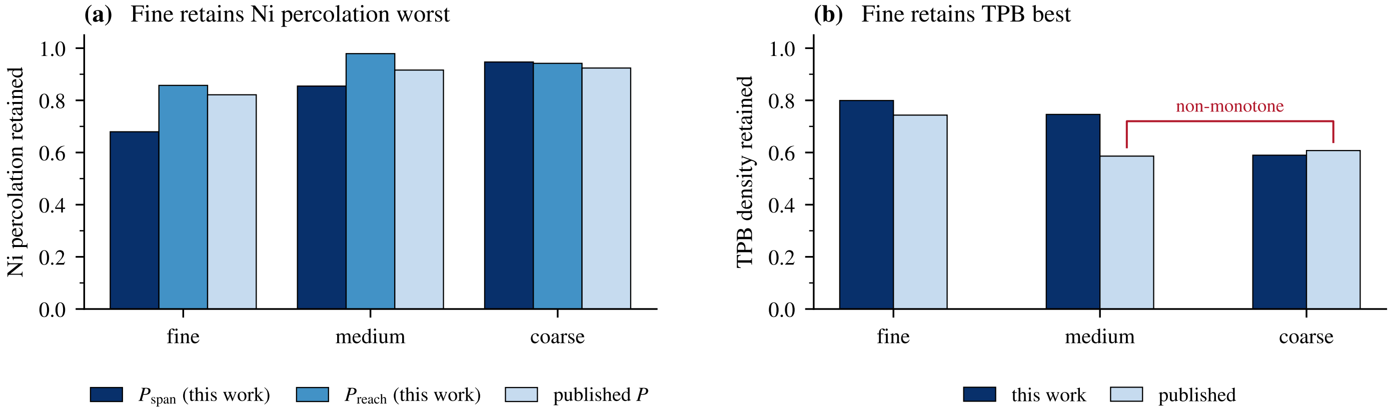

Figure 1: The measured signature the operators were built to reproduce. (a) Retained nickel percolation under three definitions. Fine is worst on all three; medium and coarse exchange places between P_{span} and P_{reach}, so that comparison is treated as unresolved. (b) Retained TPB density. Fine is best in both series, but the published series is not monotone, so only the two-level statement is used. Data: `phase6_comparison_table.csv`.

Table 1: Measured retention, degraded divided by pristine, on full stacks with no sub-sampling. Published values are from [9, 10]. Pristine and degraded are different specimens. No dispersion is quoted because none exists to quote: each entry is a single number computed once over an entire stack, not a mean over regions or seeds, so there is one measurement per anode per column. The region-to-region and seed-to-seed spreads reported elsewhere in this paper belong to the ROI-based graph metrics and to the damage operators, not to these quantities.

	Ni percolation retained			TPB retained	
anode	P_{span}	P_{reach}	published P	this work	published
fine	0.680	0.857	0.821	0.799	0.743
medium	0.855	0.979	0.916	0.746	0.586
coarse	0.947	0.942	0.924	0.590	0.608

3.2. A synthetic platform, and what it concealed

Work began on a synthetic three-phase platform: nickel spheres on a jittered regular lattice joined by explicit necks, with a particulate YSZ generator whose sintering yield sets ceramic connectivity independently of grain size. The platform passed a twelve-part qualification suite on volume fractions, percolation, particle sizes, neck sizes, length-scale ordering and ceramic fragility ordering. It was, by every check we thought to apply, a good analogue.

Three mechanisms were tested on it and none produced the target ordering. Widening the lower tail of the neck-size distribution by 1.33 and 2.00 times at conserved nickel loading moved the bisection midpoint from 8.54 rounds to 8.50 and 8.50, a change of −0.04 rounds against a pre-registered interpretability threshold of 1.0; 74 of 75 bisections returned exactly 8.5. Severing every throat in the lower quartile of the size distribution (100% of candidates, at a cost of 0.6 % to 5.3 % of the nickel volume) produced no percolation failure whatever, with $P_{span} = 1.0000$ at the maximum intensity in 15 of 15 structures. Progressive fracture of sintered YSZ contacts did separate the classes, by 2.87 rounds in the predicted direction with non-overlapping seed ranges, and a matched-fragility control in which pristine ceramic fragility was equalised across classes inverted that separation to −0.27 rounds. The apparent mechanism was pristine-state loading: the class calibrated to start nearest its percolation threshold failed first, as it must. Without the control the main-arm result would have read as a confident positive.

A graph audit run to explain the first two nulls found something more general than the nulls themselves. The minimum source–sink cut of the synthetic nickel network is exactly one complete lattice cross-section: 36, 25 and 16 edges for the three classes, which are 6^2, 5^2 and 4^2, in every one of fifteen structures, with zero seed-to-seed variance (Fig. 2). A jittered regular lattice has no locally weak plane: every cross-section is statistically identical, so the minimum cut is a plane and its size is fixed by the geometry rather than by the microstructure. Removing half of all necks in random or increasing-area order still leaves 91 % to 97 % of the nickel in the spanning cluster, while a targeted cut breaks medium at 5.8 % of throats and coarse at 7.1 %, a factor of about eight apart. Fine cannot be broken by neck removal at any budget. Jitter of 0.15 of the lattice pitch lets neighbouring spheres approach to within less than their combined radii, so 13.1 % of fine's bonds are carried by direct sphere overlap independent of their neck; severing the graph minimum cut leaves the voxel mask spanning at exactly 1.0000 in all five seeds. The graph model understates the platform's connectivity, and it understates it most for the class the study needed to be most fragile. The critical edges overlap the lower-quartile

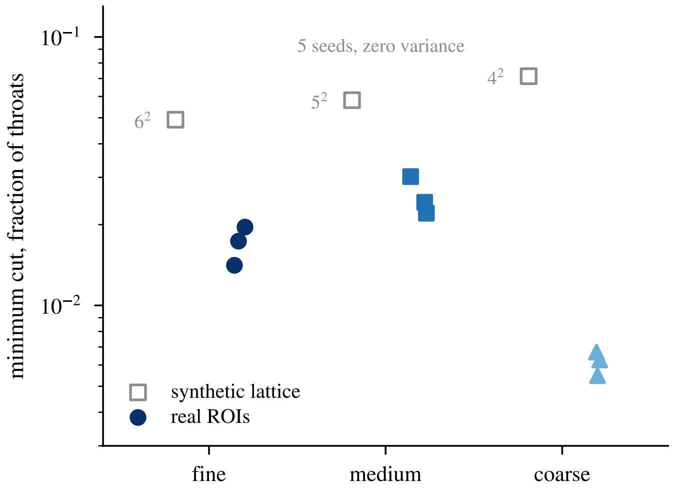

Figure 2: Minimum source–sink cut as a fraction of all throats, logarithmic axis. The synthetic lattice cuts at exactly one full cross-section, at the same fraction in all five structure seeds of each class. Real regions of interest cut at 0.55 % to 0.67 % (coarse), 1.41 % to 1.96 % (fine) and 2.21 % to 3.02 % (medium) of throats, with region-to-region spread within every anode. Data: `audit_ni_vulnerability.csv`, `c1real_results.csv`.

neck population at 0.250, 0.280 and 0.188 against a chance expectation of 0.25, which is why severing the narrow necks did nothing: in this architecture the narrow necks are not load-bearing.

The same audit measured the real networks for comparison. Real nickel networks are disordered in a way the lattice is not: connected-pair-distance coefficients of variation of 0.38 to 0.45, coordination standard deviations of 1.8 to 2.9 against a lattice whose interior coordination is fixed, and minimum cuts of a handful of edges rather than a plane. This is the first of the failure modes, and it is a statement about what a synthetic platform can be asked to do. A regular lattice cannot fail the way a real network fails, and no local stochastic operator can make it do so. Bond dilution does not help, because a diluted regular lattice remains statistically homogeneous at large scale and the cut plane merely thins; the property the lattice lacks is positional disorder, which cannot be created by deleting bonds from a periodic grid.

That conclusion retracts the three mechanism results above, and we would rather say so than leave them standing. The neck-widening and neck-severing nulls are explained by the minimum-cut measurement: an operator confined to the lower quartile of the throat distribution was attacking edges that carry no more load than any other quartile, on a network that only a globally coordinated cut can break. Those are nulls about the platform. The ceramic fracture result is disqualified independently by its own matched-fragility control. None of the three is evidence about real electrodes, and none is counted as an eliminated mechanism anywhere in this paper. What survives from the synthetic phase is the diagnosis of why it could not work.

The second failure mode surfaced only when the operators were moved to real data. Measured pristine spanning fractions per region are 0.9821/0.9754/0.9877 (fine), 0.9713/0.9446/0.9554 (medium) and 0.8878/0.9528/0.9157 (coarse). Between 1.2 % and 11.2 % of the nickel in a real electrode is already disconnected before service, and the disconnected fraction is coarseness-ordered. The published percolation values carry the same message less sharply, at 0.959 to 0.985 rather than unity. Our synthetic qualification gate required pristine P_{span} to equal 1.0000 exactly, and every structure passed it. That gate is unrepresentative of every real electrode measured here, and it is what hid the next failure mode for the entire synthetic phase of the work.

3.3. The outcome metric rewrites its own denominator

The spanning fraction divides spanning-cluster voxels by total phase voxels, which is how percolation is reported for this dataset and for electrodes generally [9]. Damage operators conventionally retain only the largest connected component, on the physical grounds that isolated nickel is electrically dead; the same reasoning underlies the standard partition of electrode microstructure into active, dead-end and isolated clusters [4]. Neither convention is ours and neither is wrong. Together they are circular. Pruning deletes exactly the non-spanning voxels, so after pruning the surviving component *is* the spanning one and P_{span} equals unity identically. The operator rewrites the denominator of the metric used to judge it. This is the third failure mode and the one that cost us the most time.

We state it as a finding rather than a slip because of how it behaved when we tried to fix it. Our first diagnosis was that an oxidative dilation step was bridging the disconnected fraction, which is physically plausible and would have made this an operator defect with an operator remedy. We removed the dilation step. The fine anode's TPB inflation moved from 15.24 to 14.80 and P_{span} remained exactly 1.0000 in all three anodes (Fig. 3a). The pathology survived a targeted, mechanistically motivated repair, because the cause was the pruning step and not the dilation.

Three properties make it hard to catch. The metric is correct on the pristine structure, where across the nine regions it returns 0.888 to 0.988 rather than unity. It is correct under any operator that does not prune. And on a test structure whose qualification gate forces pristine P_{span} to unity it is unobservable, since the pruning step has nothing to delete. A group validating on synthetic structures would pass every check either convention imposes alone and would ship the combination.

The behaviour is identical in two independently implemented operators: on real regions, pristine P_{span} of 0.9821, 0.9713 and 0.8878 becomes exactly 1.0000 at the first damage step in both.

This is a metric–operator incompatibility rather than an operator defect, and the fix belongs to the metric. Scoring on spanning-cluster voxels as a fraction of *pristine* nickel gives a quantity that does not move when isolated material

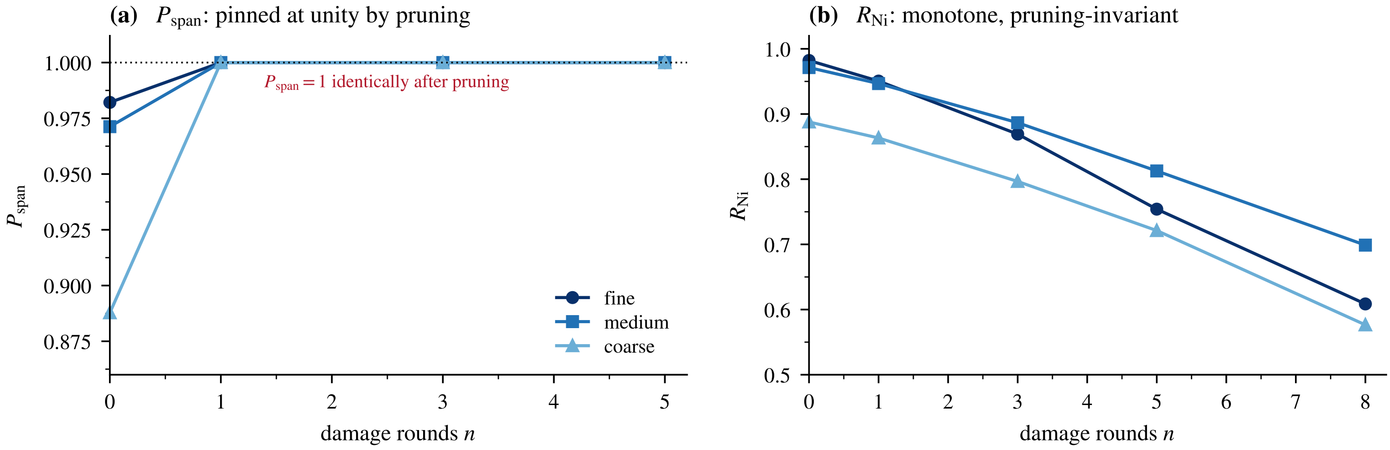

Figure 3: (a) Spanning fraction under surface erosion on real regions of interest. Every anode reaches exactly unity at the first damage round and stays there, because largest-component pruning deletes the non-spanning voxels that form the denominator. (b) The pruning-invariant metric R_{Ni} on the same runs declines monotonically from a pristine value that is not forced to unity. Data: `c1real_o6_validity.csv`, `c1real_rni_gate.csv`.

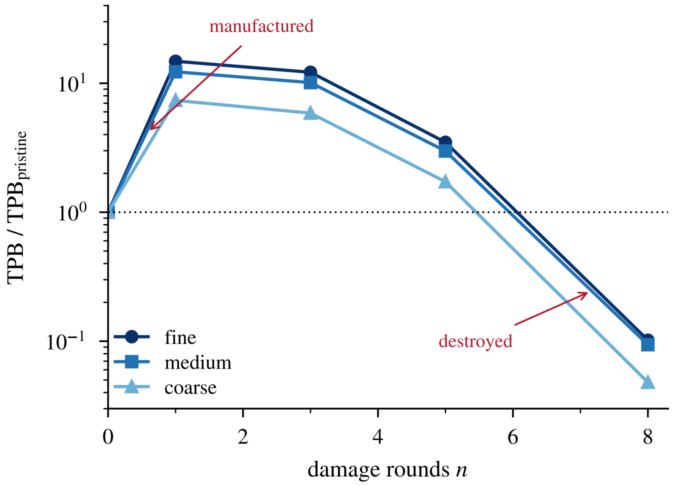

Figure 4: TPB density relative to pristine under surface erosion on real regions of interest. Stochastic single-voxel removal pits the nickel surface, and every new nickel/pore facet touching zirconia creates triple line. Any TPB conclusion drawn before the peak reports discretisation, not the electrode. Data: `c1real_rni_gate.csv`.

is deleted, falls monotonically under erosion (Fig. 3b), and is directly comparable to the measured retention values. On the synthetic platform the pathology was invisible for a precise reason: the qualification gate had forced pristine P_{span} to unity, leaving the pruning step nothing to remove.

3.4. Voxel-scale moves manufacture TPB, whatever the operator's budget

The fourth failure mode is in the same runs, and it inverts the quantity the electrode is built around before destroying it. A voxel-scale operator does not begin by removing triple line; it begins by manufacturing it, because removing single voxels roughens a surface at the grid scale and every newly exposed nickel/pore facet that happens to touch zirconia adds junction. Any TPB conclusion drawn during that excursion reports the discretisation and not the electrode.

The excursion is large. TPB density is multiplied by 14.8, 12.3 and 7.4 at the first damage round before it collapses to 4.8 % to 10.2 % of pristine by the eighth (Fig. 4).

Removal is not the cause, and neither is a loose volume budget. A volume-conserving redistribution operator run on synthetic structures showed the same signature more mildly, raising nickel surface area by 1.4 % to 5.3 % at exactly conserved volume and roughly doubling TPB, with per-class ratios of 2.02, 2.10 and 2.26. The sharper test is a single-voxel swap that conserves volume exactly *and* decreases specific surface area monotonically, run sequentially on the three real regions of interest. It multiplies TPB density by 5.0, 4.9 and 3.7 for the fine, medium and coarse anodes while satisfying both budgets throughout (`o7_gate_a1v2_real.py`). Neither conserving mass nor reducing area prevents the artifact.

The reason is that the two quantities are not sensitive to the same thing. Specific surface area is a facet count over the nickel/pore interface; TPB is a one-dimensional junction count where three phases meet. Under the swap rule, about 80 % of accepted moves carry $\Delta A = 0$ exactly—on the fine anode, 208 180 of 258 870—and are therefore unpriced by an area criterion (Fig. 8b). An area budget does not constrain the degree of freedom that governs TPB.

Where those unpriced moves land is worth measuring rather than assuming, and the measurement is more specific than the guess. The rule removes nickel from sites with few nickel neighbours, and nickel is thinnest against its own kind exactly where it meets zirconia, since a voxel on that contact is backed by ceramic rather than by more nickel. On the fine anode 92.8 % of removed voxels are adjacent to YSZ: the operator strips nickel off the ceramic. Added material, by contrast, lands almost anywhere, only 3.2 % of it beside YSZ, and arrives as 34753 fragments of median size two voxels.

Re-applying the two groups separately to the pristine structure separates their contributions (Fig. 9). The contact-adjacent moves alone reproduce the whole artifact, ×5.07 against ×5.03 for the full set; every other move, which is 97 % of the added material, gives ×1.04. The mechanism is therefore specific rather than diffuse: the area criterion rewards precisely the moves that pit the nickel–zirconia contact patch, and the triple line lives on that patch's perimeter. The gate prices the nickel/pore interface; the damage happens on the nickel/zirconia interface, which the gate never inspects.

Forbidding those moves reduces the artifact without removing it, and the residue is the more useful number. Tightening the rule to strict inequality, $\Delta A < 0$, so that every accepted move must lower the area, leaves TPB ratios of 1.53, 1.33 and 1.34 against 5.0, 4.9 and 3.7 under $\Delta A \leq 0$ (`o7_strict_inequality.py`). Most of the inflation is indeed the neutral plateau; a third to a half of a doubling is not. Nor is a matched-budget control needed to reach that conclusion. The greedy rule selects the smallest $\mathrm{nb}(a)$ against the largest $\mathrm{nb}(b)$, so the extremal pair attains the minimum ΔA available; if any strictly negative move exists, that pair is strictly negative. Neutral moves are taken only once negative ones are exhausted, so the first M moves under $\Delta A \leq 0$ are exactly the M moves the strict rule makes, and the two operators agree until the strict one halts.

The consequence is that the criterion cannot be repaired by tightening it. A strictly area-lowering operator still manufactures triple line, still raises R_{Ni}, and still never thins a neck. The problem is not that the area budget admits a null space; it is that surface area is the wrong invariant to validate a coarsening rule against, because a rule can satisfy it in full while moving the microstructure away from the physics it is meant to represent.

That voxelised TPB estimates carry a resolution-dependent systematic bias is established [14, 15]. What is different here is that the bias is not a fixed offset but a large, non-monotone excursion driven by the operator itself, so it cannot be cancelled by taking ratios to the pristine state or by any calibration performed before damage. An operator whose TPB retention exceeds unity is roughening rather than eroding, and this is worth stating as a routine check: it costs one line and it disqualified three operators here. On our synthetic platform the same excursion was inconspicuous because the platform's absolute TPB was already about eightfold too high, an artifact of synthetic phase placement rather than of the estimator. On real data the corrected estimator returns 4.48 to 4.66, 1.87 to 2.37 and $1.47\,\mu\mathrm{m}^{-2}$ to $1.55\,\mu\mathrm{m}^{-2}$ for the three anodes against published values of 3.62, 2.11 and 1.47, a factor of 1.0 to 1.3.

3.5. Surface erosion reverses the measured ordering

With a pruning-invariant metric and an operator that passes its remaining validity checks, the mechanism test can finally be run. It fails, and it fails in the wrong direction.

The claim to be tested is a two-level one, because that is all the measurement supports: the fine anode retains nickel percolation worst, and therefore should be the first to lose it. The simulation puts it last. That statement is the result; the numbers below establish it and then establish that it does not depend on where the failure threshold is placed.

Bisecting the damage intensity at which R_{Ni} falls below 0.50 gives class means of 9.72 (fine), 9.06 (medium) and 9.39 (coarse) rounds; at 0.10 the values are 9.72, 9.28 and 9.39. The pre-registered criterion required fine to be lower than both others by at least one round. It is higher at both (Fig. 5a,b).

The three-way ordering among those means should not be read, and we do not read it. The separations are a few tenths of a round against region-to-region and seed-to-seed standard deviations of 0.33 to 1.00, so which of medium and coarse fails first is not resolved, exactly as medium versus coarse is unresolved in the measurement itself.

What is resolved is the sign, and a threshold sweep shows it is not an artifact of the pre-registered cut. Recording the full R_{Ni} curve over 20 rounds for every region and seed, and reading nine thresholds from 0.80 to 0.10 off the same damage realisations, the fine anode is the first to fail at *none* of them (Fig. 6). The identity of the anode that does fail first changes with the threshold—coarse above 0.60, medium at and below 0.50—which is a further symptom of the unresolved pair, but fine is never among them. The sweep includes 0.68, the fine anode's own measured retention, which is the threshold at which a placement artifact would be most likely to show; it behaves like its neighbours.

Two incidental confirmations fall out. The five thresholds at and below 0.50 return an identical transition for the fine anode, 9.72 rounds in every case, because R_{Ni} crosses that entire range within a single damage round; those five were never five independent measurements. And the pre-registered pair of thresholds, 0.50 and 0.10, sit inside that degenerate band, so the original design had less resolution than its two-threshold form suggested.

Against the measurement, then, the operator inverts the one comparison the data supports: real retention makes fine the worst, and every simulated threshold makes it the best or the middle.

Two candidate explanations were measured directly and neither survives. A kinetic hypothesis predicts that the anode with the largest specific nickel surface area erodes fastest; the raw Spearman coefficient is +0.018 at the 0.50 threshold and +0.089 at 0.10 (Fig. 5c). A topological hypothesis predicts that the anode with the smallest minimum cut fails first; the coefficients are +0.028 and +0.213. Nothing reaches the pre-registered $|\rho| \geq 0.6$, partial correlations controlling for pristine P_{span} change no sign that matters, and leave-one-out signs are inconsistent for three of the four combinations.

The minimum-cut measurement produces a result worth separating from the null. With the coarse regions properly sized at $12\,\mu\mathrm{m}$ (258 to 299 graph nodes, against 48 to 72 in an earlier under-sized pass), the coarse anode is by a

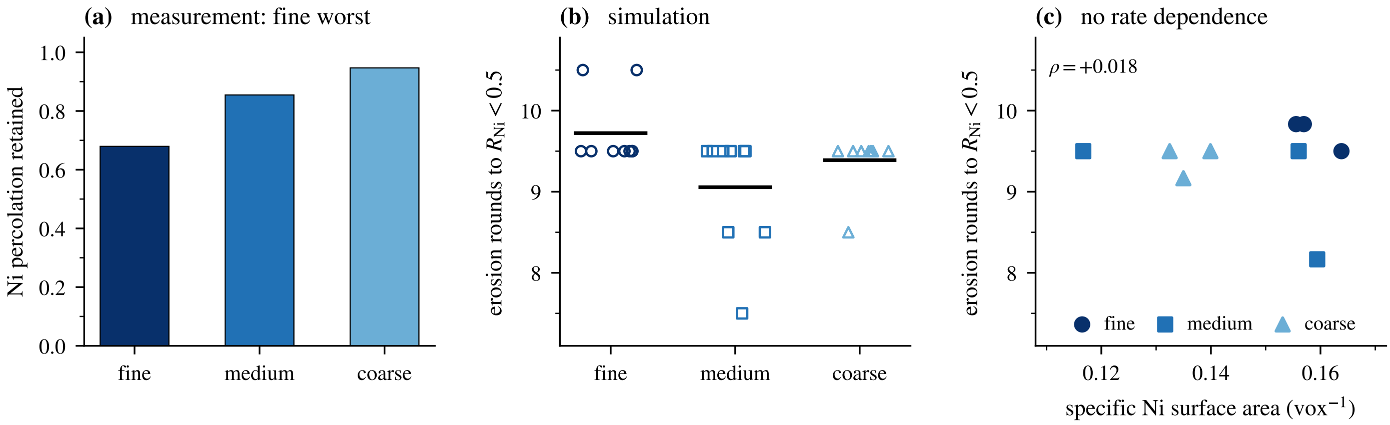

Figure 5: (a) Measured nickel percolation retention: fine is worst. (b) Erosion rounds to $R_{\mathrm{Ni}} < 0.5$ on the same real microstructures, three regions and three damage seeds per anode; points are individual bisections and the horizontal bar is the class mean. Fine is the *last* anode to fail. (c) Transition against pristine specific nickel surface area, one point per region. Data: `c1real_results.csv`, `phase6_comparison_table.csv`.

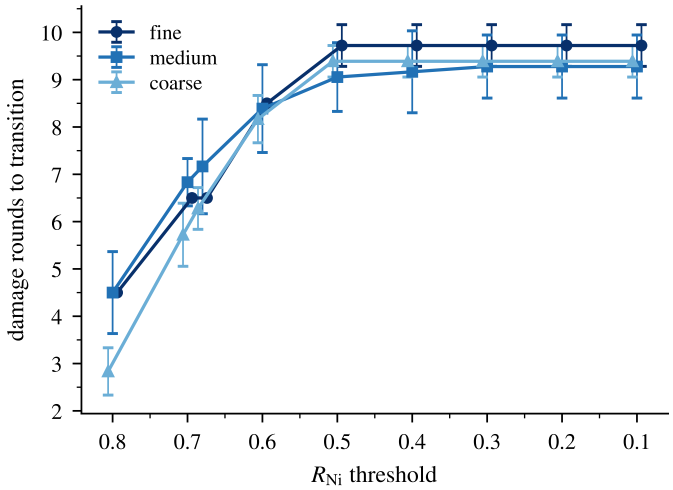

Figure 6: The reversal does not depend on where failure is declared. Damage rounds to transition against the R_{Ni} threshold, class means over three regions and three damage seeds, error bars one standard deviation. Every threshold is read from the same damage realisations, so the curves differ only in where the cut is placed. Nine thresholds are plotted; the unlabelled point between 0.70 and 0.60 is 0.68, the fine anode's own measured retention. Series are offset slightly in the threshold direction so that exact ties stay visible; fine and medium coincide at 4.50 rounds at a threshold of 0.80. The fine anode is the first to fail at none of the nine thresholds; the measurement says it should be first at all of them. The convergence below 0.50 is the discontinuous collapse of the spanning cluster: R_{Ni} crosses that whole range inside one round. Data: `o7_threshold_transitions.csv`.

wide margin the most topologically fragile of the three: minimum cut fractions are 0.0141 to 0.0196 (fine), 0.0221 to 0.0302 (medium) and 0.0055 to 0.0067 (coarse). On class means the coarse minimum-cut fraction is 2.8 times below fine and 4.1 times below medium. Coarse is also the best retainer in service, at 0.947, so pristine topological fragility runs several-fold against the outcome it is supposed to predict. Taken with the absent surface area correlation, the sharpest statement this dataset supports about the measured ordering is a statement about what does not control it: neither the pristine topology nor the specific surface area does.

Why the erosion operator behaves as it does is visible in the synthetic runs, where the volume loss at the transition was recorded. Median across 45 bisections, fine loses 74.4 % of its nickel before the network breaks, medium 70.5 % and coarse 62.8 %. Fine does erode faster, and its specific surface area really is the largest of the three on real regions, 0.159 against coarse's 0.136, but its network tolerates proportionally more volume loss before it stops spanning. High specific surface area and high connectivity redundancy are both consequences of fineness and they oppose each other in this outcome variable, cancelling almost exactly.

3.6. The area barrier

Surface erosion removes volume. Real coarsening does not: it conserves nickel and reduces surface area. A removal-only operator also cannot reproduce the measured change in nickel volume fraction, which is −0.1006, −0.0164 and +0.0145 across the three anodes and therefore *increases* in the coarse case. So the natural next operator is volume-conserving, and it is the one that produces the most general result in this paper.

For a single-voxel swap on a 6-connected lattice the change in exposed nickel area is exact rather than approximate. Removing a surface voxel a changes the count of exposed nickel faces by $2\,\mathrm{nb}(a) - 6$, where nb counts nickel neighbours, and adding a voxel at a pore site b changes it by $6 - 2\,\mathrm{nb}(b)$. Hence

$$\Delta A = 2\,[\,\mathrm{nb}(a) - \mathrm{nb}(b)\,], \tag{2}$$

so that $\Delta A \leq 0$ if and only if $\mathrm{nb}(a) \leq \mathrm{nb}(b)$. Material must leave a site with few nickel neighbours, which is convex and of high curvature, and arrive at one with many, which is concave (Fig. 7a). The expected direction of matter flow is recovered without approximation.

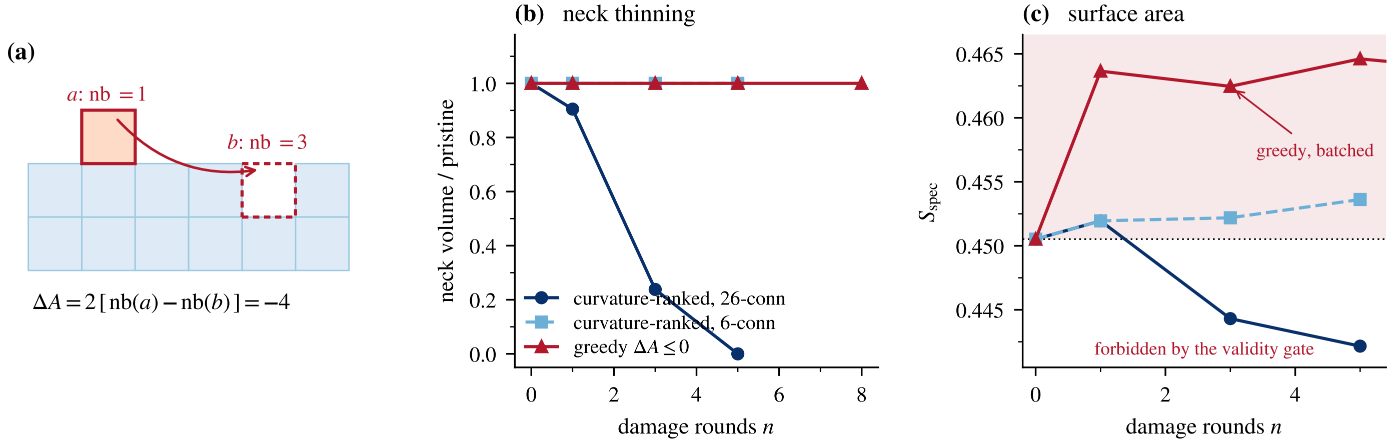

Figure 7: Two operators bracket an impossibility on a sphere–neck–sphere test body (neck 63 voxels, pristine $S_{\mathrm{spec}} = 0.45052$). (a) The exact area change of one swap on a 6-connected lattice, Eq. (2): moving a voxel from a protrusion to a notch lowers exposed area by four faces. (b) Neck volume. The curvature-ranked operator with a 26-connectivity stencil thins the neck to zero; with 6-connectivity it does nothing; the greedy operator leaves the neck intact. (c) Surface area. The curvature-ranked operator enters the region forbidden by its own validity criterion at $n = 1$, at 0.45195 against a pristine 0.45052, before falling below pristine at $n = 3$. The greedy operator accepts 96% of proposals and also rises above pristine, to 0.46365 at $n = 1$, because per-move $\Delta A \leq 0$ guarantees do not compose across a batch ranked on a stale neighbour field. Data: `o5v2_area_barrier.csv`, regenerated by `o5v2_regenerate.py`.

A coarsening operator ought to reduce specific surface area, so requiring it to do so monotonically is the criterion such a rule invites. We pre-registered exactly that gate; we do not claim to have found it imposed elsewhere, and its force here comes from being the check a careful implementer would adopt, not from being a documented convention. Two move-selection rules were implemented against it and they bracket the space of single-swap algorithms between them.

The first ranks candidate moves by a curvature proxy and applies them without testing the area change. With a 6-connectivity stencil it does nothing useful, leaving the model neck at 63 voxels; with a 26-connectivity stencil the neck thins monotonically, 63 to 57 to 15 to 0, at exactly conserved volume (Fig. 7b). This is genuine Rayleigh-type behaviour, and it fails the gate, because at the first round the surface area *rises*, to 0.45195 against a pristine 0.45052, under both stencils (Fig. 7c). Curvature rank is a proxy for the area change and not the area change itself: on a discrete lattice the added voxel exposes new faces at its own free sides, and at $n = 1$ that transient exceeds the area removed. By $n = 3$ the added material has coalesced into the neck and the net area falls below pristine. This is a fifth failure mode in its own right, and it is the one that Eq. (2) was derived to repair.

The second rule uses Eq. (2) directly, accepting a swap only when $\Delta A \leq 0$. It accepts 96% of proposals on the test body, and it still fails the gate—but not by refusing to move. Applied in batches of $k = 0.03\,N_{\mathrm{surf}}$ per round, as implemented, the specific surface area *rises* at every intensity, from 0.45052 pristine to 0.46365 at $n = 1$ and 0.46461 at $n = 5$. Equation (2) is exact for one isolated swap against a current neighbour field; a batch is ranked once against a stale field, the moves interact, and the per-move guarantees do not sum.

Removing that confound does not rescue the rule. A strictly sequential variant—one move, neighbour field recomputed, repeat—does lower the area, and on all three real electrodes it passes the strict-decrease gate: specific surface area falls by −0.00560, −0.00251 and −0.00468 from pristine values of 0.15696, 0.11670 and 0.13991 for the fine, medium and coarse anodes, that is by 2.2 % to 3.6 %, at exact volume conservation. The rule is permissive rather than selective on real data: essentially every extremal pair it examines is admissible, and it accepts until it runs out of its move budget rather than out of legal moves. It fails instead on two conditions that the synthetic test body could not probe, because that body has no nickel–zirconia contact and does not span its domain, so its triple-phase-boundary density and percolation are identically zero. On the real regions of interest, triple-phase-boundary density *rises* by factors of 5.0, 4.9 and 3.7 for the fine, medium and coarse electrodes, and the retained percolating fraction R_{Ni} rises rather than falls.

The mechanism is the area-neutral plateau. Roughly 80% of accepted moves carry $\Delta A = 0$ exactly: on the fine electrode, 208 180 of 258 870. They are free under a surface-area criterion and are therefore unconstrained, and they concentrate where nickel is thinnest against its own kind, which is the contact with zirconia; 92.8 % of removed voxels sit against YSZ, and re-applying only those moves reproduces the entire inflation. A monotone-area gate cannot detect this, because the area genuinely does decrease while the microstructure degrades in a way the criterion does not measure.

That is the whole of the argument, and its consequence is a structural one. Rayleigh-type break-up of a solid cylinder by surface diffusion is a collective, long-wavelength instability [17]: a cylinder is unstable to perturbations of wavelength greater than its circumference, but the instability only lowers energy once a correlated set of voxels has moved together, and any single-voxel perturbation raises the area first. One operator crosses that barrier without pricing it and violates the criterion; the other prices it and never crosses, spending its moves instead on the zero-cost plateau, where they redistribute nickel without correlating. In no variant, at any intensity, seed or move budget tested, did the neck thin: the mechanism the rules exist to represent was never produced. A monotonicity gate forbids the only bridge, finite-temperature acceptance of area-increasing moves. Any single-voxel-swap coarsening study that both enforces monotone surface-area reduction and claims to reproduce Rayleigh-type neck break-up is therefore either not producing that break-up or is violating its own validity criterion. The two cannot hold together. This is the sixth and most general of the failure modes, and unlike the other five it is a statement about what an operator class can represent rather than about how a particular implementation goes wrong.

Its generality does not rest on the test body. Equation (2) is exact on any 6-connected lattice, and the local-minimum argument follows from two facts of geometry rather than from this dumbbell: a sphere minimises area at fixed volume, and a straight cylinder is stable against the displacement of any individual voxel. The runs in Fig. 7 demonstrate the consequence on a case where the intended mechanism is unambiguous; they are not what establishes it. The scope is correspondingly specific. The argument covers operators that move one voxel at a time. It says nothing about morphological openings and closings, block moves, or any update that displaces a correlated set of voxels together, and a correlated update is precisely what could cross the barrier.

What concealed this on the synthetic platform is simple: the operator was never asked to thin a neck against an explicit area budget, because the structures were fully connected and the validation was about connectivity. We stopped at this point, as the pre-registration required. The agglomeration hypothesis is not falsified by any of this. It is untested, because the operator class available to us cannot represent it under a gate we still believe is the right gate for that operator class.

3.7. Two hypotheses these data cannot pose

Two further hypotheses were pre-registered with explicit untestability branches, and both branches fired. We report them because a pre-registered untestable outcome is information, and because the alternative—redefining the criterion until a number emerges—is the failure mode this paper is about.

The first is redistribution: nickel migrating from the electrolyte-adjacent functional layer into the support, most strongly in the finest microstructure. Testing it requires an electrolyte interface to be present in both stacks of a pair. The criterion, a sustained gradient of at least 0.05 in absolute zirconia volume fraction between the two ends of a stack, is met in one pair of three, and not in the fine anode, which is the anode the hypothesis concerns: gradients are 0.0042 and 0.0240 for fine in its two states, 0.0059 and 0.0552 for medium, and 0.1035 and 0.1157 for coarse. Nickel content does vary strongly with depth in every stack, with coarse post-redox spanning 0.045 to 0.494 across 0.5 µm slabs, but in fine and in pristine medium that variation is not organised as a monotone interface gradient. It is spatial heterogeneity, which is a different thing. The likeliest explanation is prosaic. These subvolumes are 19 µm to 24 µm across the through-thickness direction against a support hundreds of micrometres thick, and there is no reason a given tomogram must straddle the interface.

The second is cumulative-strain fracture of the ceramic scaffold, motivated by the measured YSZ retention of 0.958, 0.865 and 0.000, in which the coarse anode loses ceramic percolation outright. A deterministic operator broke YSZ throats whose local redox strain, accumulated over cycles, exceeded a threshold frozen before implementation. Both declared untestable conditions fired. The coarse anode, which is the anode the hypothesis is about, yielded no extractable YSZ throat graph at all: at a 12 µm region the YSZ phase does not span the transport axis, which is itself consistent with coarse YSZ being the most fragmented phase in the dataset at a full-stack pristine P_{span} of 0.9246. And in the two anodes that did yield graphs the operator is nearly inert, breaking 1.0 % to 1.2 % of throats at the maximum cycle count and moving YSZ retention to 0.9995 and 0.9996 against targets of 0.958 and 0.865.

The diagnosis is a design lesson about pre-registration rather than about nickel. The strain proxy is bounded above by the frozen strain scale $\varepsilon_0 = 0.0065$, and after smoothing the per-throat values peak at 0.00413 and 0.00210. The frozen threshold of 0.010 sits above the maximum single-cycle strain by factors of 2.4 and 4.8, so only cumulative amplification lifts any throat over it. We froze an absolute threshold against a quantity whose scale we had not yet measured. The remedy is to measure the strain distribution on one region first and pre-register the threshold as a quantile of it. We did not re-pick the threshold after seeing the distribution, because doing so would be fitting a threshold to produce damage. Independent support for the strain scale itself does exist. Reviews of redox cycling report approximately 1 % linear expansion on the first complete oxidation of a Ni/YSZ composite [2], against the 0.65 % we froze. The strain scale was therefore not the error, though that does not rescue the run.

4. Discussion

The six failure modes share a structure worth naming, set out in Table 2. In each case a modelling convention that is individually reasonable becomes wrong in combination with another, and the combination is invisible on the synthetic structures normally used for validation. Retaining

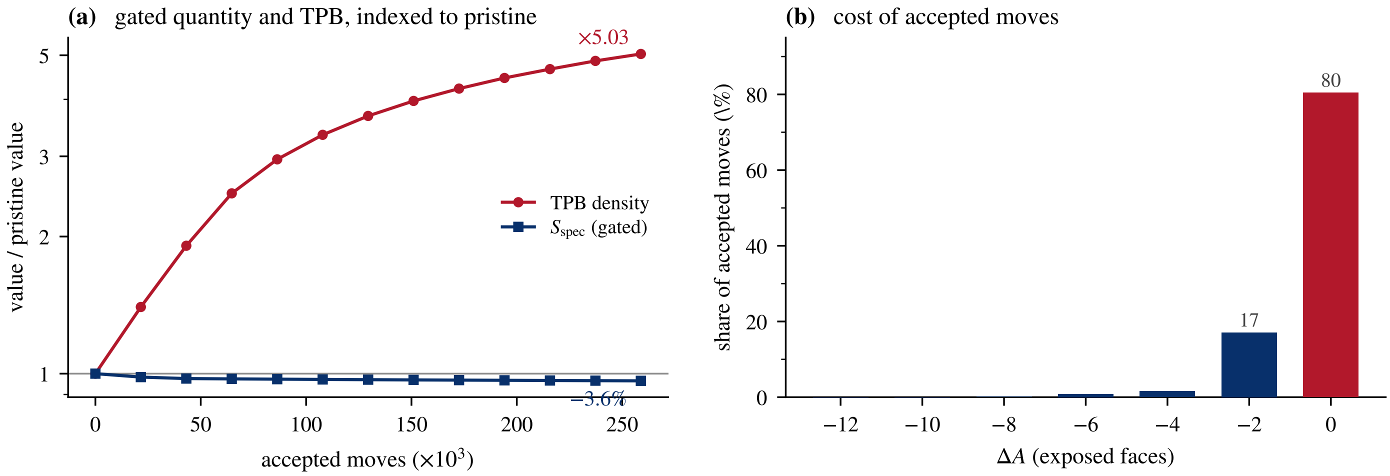

Figure 8: The validity criterion is satisfied while the quantity of interest is destroyed. Fine anode, 67 million voxels, sequential area-decreasing swap, volume conserved exactly. (a) Both quantities are divided by their pristine values so that they share one axis: specific surface area, the gated quantity, falls 3.6 %, while triple-phase-boundary density rises by a factor of 5.03. (b) Why the gate does not object. Accepted moves by their cost in exposed faces; four in five carry $\Delta A = 0$ exactly and are unpriced by the criterion. Data: `o7_trajectory.csv`, `o7_da_histogram.csv`.

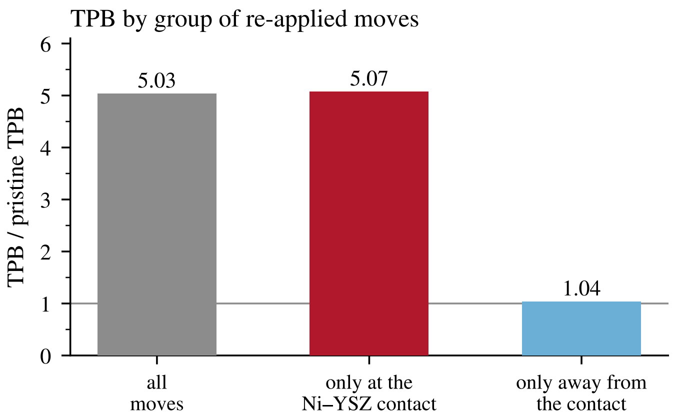

Figure 9: Where the triple-line inflation comes from, by re-applying the two groups of moves separately to the pristine structure. Moves that touch the nickel–zirconia contact reproduce the whole artifact on their own; every other move, 97 % of the added material, changes TPB by four percent. Data: `o7_counterfactual.csv`.

only the largest connected component is reasonable; measuring the spanning fraction against total phase volume is reasonable; together they are circular. Requiring perfect pristine connectivity of a synthetic analogue is reasonable; it also removes the condition under which that circularity would have been visible. Validating a coarsening operator by monotone area reduction is reasonable; so is expecting it to reproduce capillary break-up; together they are contradictory. A regular lattice is a reasonable test structure; a stochastic local operator is a reasonable model of degradation; together they cannot produce a percolation failure, because the lattice's minimum cut is a plane and local action cannot find one.

The prescription that follows is not a general appeal to rigour but something narrower and cheaper. Validation criteria and mechanisms should be checked against each other explicitly, and preferably on structures that are imperfect in the way real ones are. Three of the six modes above would have been caught by a test structure that was merely *not* fully connected, and a fourth by asking whether the operator can in principle produce the move the mechanism requires. A synthetic object that is fully connected, periodic and smooth will pass suites that a real microstructure fails, and it will conceal precisely the interactions that matter.

4.1. Relation to continuum and discrete prior work

The impossibility of Section 3.6 is specific to voxel-swap operators validated by monotone area reduction, and it does not extend to continuum formulations. A phase-field model evolves a continuous order parameter under a free-energy functional and is not restricted to single-voxel moves, so it can pass through the transient area increase that a discrete swap cannot. This is not a hypothetical alternative for this material: phase-field simulations of Ni coarsening have been run on three-dimensional Ni–YSZ reconstructions from ptychographic nano-tomography with measured thermophysical inputs [6], validated against ex-situ imaging of annealed specimens [5], and coupled to performance models on FIB-SEM reconstructions [7], in a line going back to the three-phase simulations of Chen and co-workers [4]. That body of work is the appropriate point of comparison, and it sharpens the present claim rather than weakening it. Where a study needs coarsening, a continuum formulation is the sound choice. Where a study uses discrete voxel operators, as many connectivity and TPB studies do because they act directly on the segmented label field, the compatibility check of Eq. (2) should be run before any validity gate is adopted.

Table 2: The six failure modes, grouped by kind rather than by the order the work encountered them. Modes 1, 2, 3 and 5 are implementation faults: each arises from a combination of individually defensible choices and each is removable by choosing differently. Modes 4 and 6 are not. They are properties of what a voxel-scale move can do at a three-phase junction, and of what a monotone-area criterion admits, so they persist under any implementation of the move class. Numbering is retained from the text. Every mode was invisible on the synthetic platform for the reason given in the last column.

	failure mode	what exposed it	what concealed it
Implementation faults: individually reasonable conventions that go wrong in combination			
1	lattice minimum cut is a plane	cut is exactly 6^2, 5^2, 4^2 edges, zero variance	regular lattices were never asked to fail locally
2	gate demands perfect pristine connectivity	real electrodes carry 1.2 % to 11.2 % isolated Ni	synthetic structures were built fully connected
3	pruning fixes P_{span} at unity	P_{span} jumps to 1.0000 at the first damage round	pristine P_{span} was already 1.0000 by gate
5	curvature rank ≠ area change	S_{spec} rises at $n = 1$ under both stencils	no explicit area budget was ever imposed
Structural limits: properties of the move class itself, which no implementation can remove			
4	voxel moves manufacture TPB	7–15× under erosion; 3.7–5.0× under a volume- and area-budgeted swap	platform TPB was already $\sim 8\times$ inflated
6	area monotonicity excludes break-up	80% of accepted moves are area-neutral; neck never thins	the operator was never asked to thin a neck

Discrete lattice methods are not thereby disqualified for this material either, and the clearest evidence is a model that makes the same move we do. A Potts kinetic Monte Carlo treatment of NiO–YSZ sintering, calibrated and validated against FIB-SEM microstructure [8], represents surface diffusion as the exchange of a pore site with a neighbouring solid site: a single-voxel swap on a voxel grid, which is our move class on our discretisation. It reaches the coarsening behaviour our operator could not. The difference is the acceptance rule. That model accepts a swap that raises the system energy with probability $\exp(-\Delta E/k_{\mathrm{B}}T)$ at a simulation temperature, so the barrier of Eq. (2) is crossed rather than avoided.

The distinction that matters is therefore not continuum against discrete, and not even swap against non-swap. It is whether energy-raising moves are permitted at all. A monotone area-reduction gate forbids them, and it forbids them as a validity criterion rather than as a choice of dynamics. That is why the impossibility is narrow, and it is also why it is easy to walk into: the gate looks like rigour and functions like a constraint on the physics. The same gate imposed on the model of [8] would forbid the energy-raising moves its Metropolis rule depends on; that follows from the definition of the gate rather than from any run of ours, and no such gate is imposed there.

The comparison should be pressed past the sixth mode, because it does not stop there. Of the six in Table 2, that model is structurally exposed to one. Mode 4 as generalised above is a property of voxel-scale moves at a three-phase junction rather than of any particular acceptance rule, and a Potts model on a voxel grid makes voxel-scale moves; it is therefore subject to the same transient inflation of triple line, and a study using one should check that its TPB trajectory does not exceed unity before reading anything from it. What that model does avoid is the persistent pathologies. It takes real reconstructions as input, so there is no jittered lattice with a planar minimum cut and no qualification gate demanding perfect pristine connectivity. It scores on volume fraction, specific surface area, tortuosity and TPB density rather than on a spanning fraction, so the circularity of the third mode has nothing to act on. Its evolution is driven by an energy functional rather than by stochastic voxel removal, so the roughening of the fourth does not arise, and it evaluates that energy explicitly rather than through a curvature proxy, which is the fifth. Three choices account for all six: real input, an explicit energy, and finite-temperature acceptance. We made none of them.

We take that as the strongest available support for the prescription of this section rather than as an argument against it. The six modes are not exotic. They are what a study meets when it declines one or more of those three choices, and a reader is entitled to conclude that the established formulation should have been the starting point. What this paper adds is not the discovery that a better method exists but the measurement of what the worse one costs. Nothing in [8] reports the effect of an area-monotonicity gate on a swap operator, because no such gate is imposed there. Equation (2), the area-neutral plateau and the bracketing pair of operators are a quantitative account of a trap that a well-designed study walks around without comment, and the value of documenting a trap does not depend on its being unavoidable.

Two of our six modes have partial precedent and we do not claim them as new. The systematic error in TPB length computed from voxelised data is documented, both as a function of calculation rule [14] and of imaging resolution [15]; what we add is that a damage operator turns that static bias into a large non-monotone excursion that no pre-damage calibration removes. That real electrodes contain isolated nickel is likewise known, and appears in the source papers for this dataset as pristine percolation values below unity [9]; what we add is the consequence, that a qualification gate demanding perfect pristine connectivity disqualifies the synthetic analogue as a place to discover metric pathologies. The remaining four (the pruning circularity, lattice minimum-cut planarity, the curvature-proxy failure and the area barrier) we have not found stated elsewhere, which is a statement about our search rather than a claim of priority.

4.2. What the negative result does and does not establish

One operator class reached a mechanism verdict on real microstructures and reversed the measured ordering. The three synthetic mechanism tests are withdrawn with the platform that carried them. We have not shown that no geometric mechanism reproduces the ordering, and on the strength of one operator class on real data we could not. We have specifically not shown that agglomeration is not the physical cause of nickel percolation loss: Section 3.6 says we could not test agglomeration with this operator class under this gate, which is a statement about the method and not about the material. Nor is anything here evidence against continuum or phase-field treatments, which are subject to none of the six modes.

The limitations are the ordinary ones and they bind hard. There are three anodes, three regions of interest per anode and three damage seeds, so class comparisons are correspondingly weak, although the sign reversal in Fig. 5 is consistent across all nine regions. Pristine and degraded stacks are different specimens, so every retention magnitude carries specimen variation and no paired quantity is available. That is a property of this dataset rather than of the problem: ex-situ tomography of a single specimen imaged before and after annealing has been done on this material [5], and it is what a repeat of this work should be built on. The two R_{Ni} thresholds are not independent, because the spanning cluster vanishes discontinuously and R_{Ni} crosses 0.50 and 0.10 within one round in most regions, which is itself a finding about how percolation fails in these structures. Regions are single-scale at 8 or 12 µm, the coarse anode being bounded above by a stack that is only 13.3 µm deep. Graph metrics use frozen watershed parameters. Coarse YSZ throats have a tenth-percentile inscribed diameter of roughly two voxels, below this project's four-voxel resolution floor, so any throat-targeted ceramic fracture rests partly on sub-resolution features.

Four further limitations are specific enough to name. First, the region sizes were set by the nickel graph and not by the ceramic. The 12 µm coarse region was chosen to clear a 150-node floor for the nickel network, and the same choice over-resolves the YSZ phase, whose contacts are far finer; the anode where a ceramic-fracture mechanism would matter most is therefore the one where our sampling suits it least. The sizes also happen to hold the voxel count near 400^3 across the three anodes, which was a computational convenience and not a representativeness argument. Second, the synthetic generator was recalibrated once against the measured connectivity of its own contact graph, which is an internal consistency check rather than a comparison with FIB-SEM minimum cuts or triple-phase-boundary distributions; the platform was never shown to reproduce those external quantities, which is part of why its results did not survive contact with real data. Third, connectivity is evaluated throughout at 6-connectivity. That is the conservative convention, but it ignores face-diagonal contacts, and the effect is not neutral across the series: the finest anode has the most surface per unit volume and therefore the most opportunity for such contacts, so its pristine connectivity is the most likely to be understated. That reasoning is correct in direction and negligible in size. Recomputing the pristine spanning fraction under 26-connectivity on all nine regions moves the fine anode by +0.0004 on the class mean and the medium and coarse anodes by nothing at all, to four decimal places (`o7_connectivity_check.csv`). The disconnected nickel fraction is unchanged at 1.8 %, 4.3 % and 8.1 %. The convention favours the fine anode as predicted and by an amount that changes no statement in this paper. Fourth, regions are cut with free, non-periodic boundaries, which truncate particles at the faces. That penalty also falls unevenly, bearing hardest on the anode whose particles are smallest relative to the region, again the fine one. Both conventions push in the same direction on the same sample, and both are frozen rather than tuned, but neither was varied.

The open question is not which operator to try next. It is what class of mechanism could produce loss of long-range nickel connectivity together with retention of most of the reaction sites, which is what 80 % TPB retention at 68 % percolation retention means in the fine anode. Uniform removal thins everything at once and eliminates both together; our synthetic runs destroy 99.5 % of TPB by the time nickel percolation is lost, some 164 times more than the real anode does. Dewetting and retraction of nickel into compact particles would break long-range connectivity while comparatively preserving the nickel/zirconia contact perimeter where TPB lives. We record it as an untested possibility rather than a surviving candidate: nothing here eliminates its competitors, and no run in this work either supports or excludes it.

Two claims must be separated here, because only one of them fails. That a greedy operator is frozen, and therefore cannot reach the mechanism, is withdrawn: corrected, it accepts almost every move. The impossibility above is untouched, because it turns on which moves are *permitted*, not on how many are taken. No single-voxel-swap operator respecting monotone area reduction thinned a neck at any intensity, seed, move budget or acceptance rule tried, strictly area-lowering or finite-temperature. The one variant that did thin it—curvature ranking under a 26-connectivity stencil, which reached zero neck volume by $n = 5$—violated the area criterion at its first step, at 0.45195 against a pristine 0.45052. Thinning and the criterion were never observed together.

Two qualifications keep that honest. Lifting the restriction is necessary but not sufficient: the finite-temperature runs permit area-increasing moves and still produce surface roughening rather than break-up. And the impossibility is analytic only for a strictly decreasing criterion. Under the non-strict form the $\Delta A = 0$ plateau is admissible and large, so a monotone rule is free to wander it; that such wandering does not reach break-up is established here empirically, over some 2.6×10^5 accepted moves, rather than proved.

The practical consequence stands. If the anticorrelation is to be explained by Rayleigh-type break-up, testing that explanation requires a formulation not restricted to monotone area reduction. That is a necessary condition on the method, not a recipe for the mechanism, and the anticorrelation itself remains an open puzzle this class of operator cannot pose.

5. Conclusions

1. Enforcing monotone surface-area reduction as a validity criterion structurally excludes Rayleigh-type neck break-up. On a 6-connected lattice the criterion reduces to $\text{nb}(a) \leq \text{nb}(b)$, which leaves a large area-neutral plateau: roughly 80% of accepted moves carry $\Delta A = 0$, are unpriced by the criterion, and redistribute nickel without correlating into an instability. A greedy area-decreasing operator therefore accepts almost every move, never thins a neck, and—on real electrodes—inflates TPB density four- to five-fold while satisfying the area criterion it is validated against. The exclusion is analytic for a strictly decreasing criterion, where no admissible move can go uphill; under the non-strict form it rests on the plateau argument together with the measurement, and is stated at that strength. The criterion and the mechanism were never honoured together by any single-swap operator tested here.
2. Five further failure modes were measured, each invisible on synthetic test structures: largest-component pruning makes the spanning fraction identically unity; voxel-scale moves multiply TPB density by 7 to 15 under erosion and by 3.7 to 5.0 under a swap that conserves volume and reduces area, so neither budget protects the estimate; qualification gates demanding perfect pristine connectivity are unrepresentative of electrodes carrying 1.2 % to 11.2 % disconnected nickel; a jittered regular lattice cuts at exactly one full cross-section with zero variance, whereas real networks cut at 0.5 % to 3 % of throats; and curvature-ranked moves do not deliver the area reduction assumed of them.
3. Stochastic surface erosion applied to real microstructures with a pruning-invariant metric reverses the measured degradation ordering: the fine anode is the last to lose percolation rather than the first.
4. Under the uniform surface-erosion operator tested here, neither pristine topology nor specific nickel surface area predicts the measured ordering. The coarse anode carries the smallest minimum cut of the three, 2.8 times below fine and 4.1 times below medium on class means, and is the best retainer in service. The scope of that statement is the operator, not degradation in general: a rule with a different spatial bias could rank them differently, and none was tested.
5. Real nickel networks fail at 0.5 % to 3 % of throats through small, region-variable, non-planar cuts, against a lattice that fails at 4.9 % to 7.1 % and always through a full cross-section. A synthetic generator intended to reproduce real behaviour must carry comparable positional disorder: coordination standard deviation 1.8 to 2.9 and a connected-pair-distance coefficient of variation of 0.38 to 0.45.

Data and code availability

All code, pre-registration documents, run reports and result tables are in the project repository, `https://github.com/mehtaarnav/voxel-damage-operators-niysz`. The tomography data itself is third-party and is not redistributed there; it is the Holzer/Pecho Ni–YSZ FIB-tomography dataset at `https://zenodo.org/records/4056538`, fetched by the download script into a directory that is deliberately not committed. Every number in this paper traces to a committed CSV: the measured signature to `out/phase6/phase6_comparison_table.csv`; the erosion bisections, minimum cuts and Spearman inputs to `out/project2/c1real_results.csv`; the metric and TPB behaviour to `c1real_o6_validity.csv` and `c1real_rni_gate.csv`; the lattice audit to `audit_ni_vulnerability.csv`; the synthetic mechanism tests to `out/platform_v2/p10_group_experiment.csv` and `out/project2/step2_02_ceiling_check.csv`, `step2_03_main.csv`, `step2_03_control.csv`; the synthetic volume loss at the erosion transition to `step2_01_main.csv` and the redistribution operator's surface and TPB ratios to `step2_05_ceiling_check.csv`; the volume-conserving gate to `o5v2_area_barrier.csv`; the redistribution and ceramic-fracture branches to `h1_depth_profiles.csv`, `h1_layer_summary.csv` and `csfm_results.csv`; and the implementation-validation constants to `validation_constants.csv`. The last two of those are transcriptions, by `scripts/project2/o5v2_transcribe.py` and `transcribe_validation.py`, of values that their runs reported in markdown rather than in a table; each row carries the report it came from. Figures regenerate with `python scripts/project2/make_figures.py`, and file names match the figure numbers.

Each pre-registration was committed before the run it governed and the chain of commits is recorded in the repository. Corrections made against our own earlier results during the work—a TPB estimator that over-counted four-fold through periodic wrap, reversed axis extents that made one audit's cut fractions provisional until re-run, a validity check that never compared against the pristine state and printed a pass on a failing run, a pre-registered secondary-outcome rule that selected a saturating measurement point, an internally contradictory plan specifying both Metropolis acceptance and strict area monotonicity, and—the one that changed a headline result—a neighbour count taken with a structuring element that includes the centre voxel, which biased a comparison across two voxel populations by one and made an operator that accepts 96 % of proposals appear to accept none—are logged in the run reports at the point where each was found. Two further specifications were found to be load-bearing rather than incidental and are now part of the operator definition rather than of its implementation: sequential rather than batched application, and random rather than insertion-order tie-breaking among moves of equal ΔA.

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